

PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Software Engineering Mini-Project

Software Requirements Specification

Online Internship and Placement Preparation Portal

Project: Online Internship and Placement Preparation Portal

Group: Team 16

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Revision History

Version	Date	Author	Change Summary
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1. Introduction

1.1 Purpose

This document specifies the software requirements of the Online Internship and Placement Preparation Portal (the Portal), version 1.0. It covers the complete product: internship listing and application management, aptitude test preparation, and progress tracking with analytics.

1.2 Intended Audience

Developers (design and implementation), testers (Software Test Plan and test cases), project team members and course instructors (review and evaluation). Section 2 gives the overall description, Section 3 the interfaces, Section 4 the analysis models, Section 5 the functional requirements by feature, Section 6 the non-functional and security requirements, and the appendices the glossary, field layouts and traceability matrix.

1.3 Product Scope

The Portal is a web application that brings internship discovery, application tracking and placement preparation into one place. Students find and apply to internships and practise aptitude tests while tracking progress; Recruiters post listings and review applicants; Admins (placement officers) moderate content and monitor outcomes.

- **Objective:** reduce the effort of finding and applying to internships and make placement preparation measurable.
- **Benefit:** students see their strengths and gaps through analytics; recruiters reach verified candidates; the placement cell gets consolidated reports.
- **Goal:** deliver a MERN-based portal with a Python analytics service as the semester mini-project.

1.4 Definitions, Acronyms and Abbreviations

See Appendix A.

1.5 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.
- ISO/IEC/IEEE 29148:2018, Requirements engineering.
- SE Mini-Project Deliverables Part-1, PES University, 2026.
- SE Mini-Project Guidelines, PES University, 2026.
- OWASP Application Security Verification Standard (ASVS) 4.0.

2. Overall Description

2.1 Product Perspective

The Portal is a new, self-contained web product. It consists of a React single-page application, a Node.js/Express REST API backed by MongoDB, and an internal Python (FastAPI) analytics service. The

only external system is an SMTP email provider used for notifications. The architecture is detailed in the Software Architecture and Design Specification.

2.2 Product Functions

- Registration, login and profile/resume management for Students and Recruiters.
- Internship listing creation, moderation, search and filtering.
- Internship application submission, review and status tracking with notifications.
- Timed aptitude tests with automatic grading and attempt review.
- Progress dashboard, practice recommendations and aggregate analytics.
- Administration: user management, question bank, reports and audit log.

2.3 User Classes and Characteristics

User Class	Description	Frequency / Expertise	Importance
Student	College students seeking internships and preparing for placements. Uses search, apply, tests, dashboard.	Frequent; basic web skills	Highest
Recruiter	Company representatives who post listings and review applicants.	Occasional; basic web skills	High
Admin (Placement Officer)	Staff who approve listings, manage tests/users and view reports.	Daily; moderate skills	High

2.4 Operating Environment

- Client: latest two versions of Chrome, Firefox and Edge on desktop and mobile.
- Server: Linux, Node.js 20 LTS, Python 3.11, MongoDB 7.
- Deployment: cloud VM or container platform (Docker).

2.5 Design and Implementation Constraints

- Technology: MERN stack (MongoDB, Express, React, Node.js) and Python for analytics, as assigned.
- Security: compliance with the security requirements in Section 6.3.
- Academic: development within one semester by a team of three; free-tier or open-source tools only.
- Resume files limited to PDF, at most 5 MB.

2.6 Assumptions and Dependencies

- Users have a modern browser and Internet access.
- An SMTP provider is available for email notifications.
- Question bank content is prepared by the team or Admin before test launch.
- Recruiter accounts are genuine; listings are still moderated by Admin.

3. External Interface Requirements

3.1 User Interfaces

- Web UI built with React; responsive layout from 360 px width.

- Role-specific navigation: Student (Search, Applications, Tests, Dashboard), Recruiter (Listings, Applicants), Admin (Approvals, Tests, Users, Reports).
- Standard header with profile menu and notification bell on every screen; consistent inline validation messages and error banners.
- Test screen shows a countdown timer and question navigator.

3.2 Hardware Interfaces

No dedicated hardware interfaces. The Portal runs on standard client devices and server infrastructure.

3.3 Software Interfaces

Component	Interface	Purpose
MongoDB 7	Mongoose ODM over TCP	Persistent storage of users, listings, applications, tests, attempts, audit logs
Analytics Service (Python, FastAPI)	Internal REST/JSON over HTTP	Computes score trends, weak categories and aggregate statistics
SMTP provider	SMTP over TLS	Sends status-change emails
Resume file store	Local/object storage API	Stores uploaded resume PDFs

3.3 Communications Interfaces

- HTTPS (TLS 1.2+) between browser and API; JSON message format.
- REST over HTTP on the private network between API and analytics service.
- SMTP with STARTTLS for email.

4. Analysis Models

4.1 Use Case Diagram

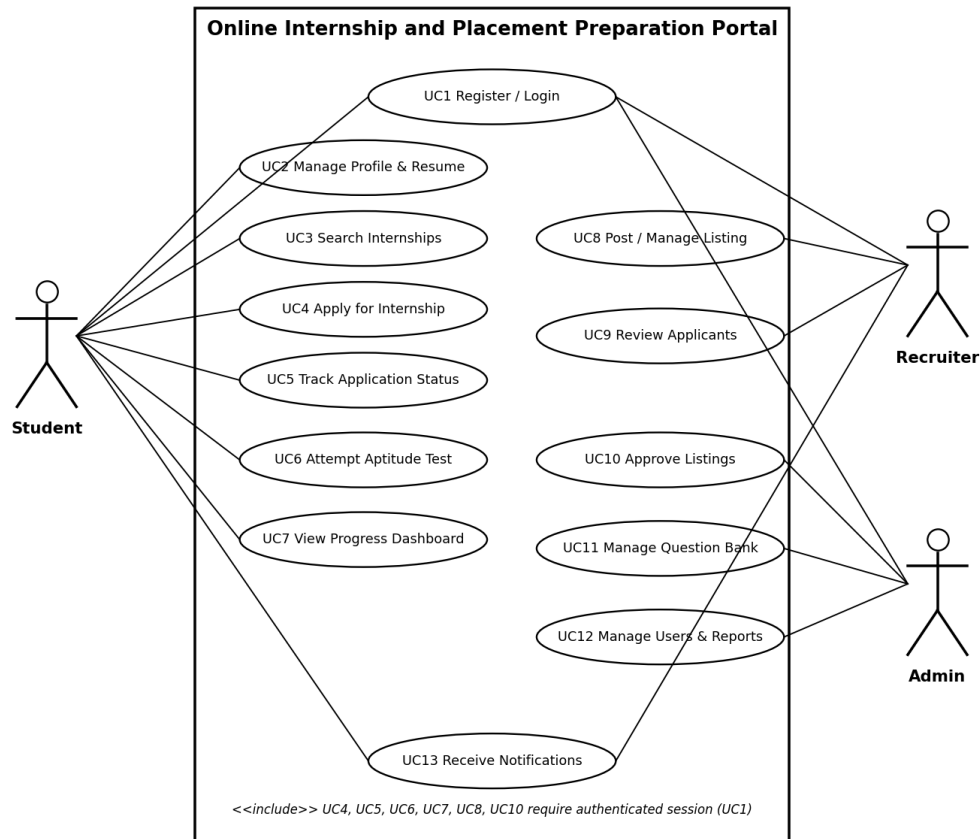


Figure 1: UML use case diagram of the Portal

4.2 Use Case Summary

Use Case	Actor	Related Requirements
UC1 Register / Login	Student, Recruiter, Admin	PIP-F-001, PIP-F-002
UC2 Manage Profile & Resume	Student	PIP-F-003
UC3 Search Internships	Student	PIP-F-007
UC4 Apply for Internship	Student	PIP-F-008
UC5 Track Application Status	Student	PIP-F-009, PIP-F-011
UC6 Attempt Aptitude Test	Student	PIP-F-013 to PIP-F-016
UC7 View Progress Dashboard	Student	PIP-F-017, PIP-F-018
UC8 Post / Manage Listing	Recruiter	PIP-F-004, PIP-F-006
UC9 Review Applicants	Recruiter	PIP-F-010
UC10 Approve Listings	Admin	PIP-F-005
UC11 Manage Question Bank	Admin	PIP-F-012
UC12 Manage Users & Reports	Admin	PIP-F-019 to PIP-F-022
UC13 Receive Notifications	Student, Recruiter	PIP-F-011

4.3 Data Model (Entities)

Main entities: User, StudentProfile, Listing, Application, Test, Question, Attempt, Notification, AuditLog.
A Listing is created by one Recruiter and has many Applications; an Application belongs to one Student and one Listing; a Test has many Questions; an Attempt belongs to one Student and one Test.

5. System Features

5.1 Authentication and Profile Management

Description and Priority: Allows users to register, log in securely and maintain a profile with a resume.
Priority: High.

Stimulus/Response Sequences:

- User submits registration form -> system validates and creates account.
- User submits credentials -> system issues token or shows error / lockout message.
- Student uploads resume -> system validates type and size and stores it.

Functional Requirements:

ID	Requirement	Priority
PIP-F-001	The system shall allow a visitor to register as a Student or Recruiter by providing name, email and a password of at least 8 characters containing at least one letter and one digit. If the email already exists or the password is invalid, the system shall reject the request and display a specific error message.	High
PIP-F-002	The system shall authenticate users using email and password and issue a session token valid for 60 minutes. After 5 consecutive failed login attempts the account shall be locked for 15 minutes.	High
PIP-F-003	The system shall allow a Student to create and edit a profile containing education, skills, and a resume in PDF format of at most 5 MB. Files of other types or larger size shall be rejected with an error message.	Medium

5.2 Internship Listings

Description and Priority: Recruiters publish listings which are moderated by Admin and discovered by Students. Priority: High.

Stimulus/Response Sequences:

- Recruiter submits listing -> status Pending.
- Admin approves/rejects -> listing becomes visible or is rejected with remark.
- Student enters search/filters -> system returns paginated results.

Functional Requirements:

ID	Requirement	Priority
PIP-F-004	The system shall allow a Recruiter to create an internship listing with title, company, description, required skills, stipend, location, mode (remote/onsite/hybrid) and application deadline. A new listing shall have the status Pending.	High
PIP-F-005	The system shall allow an Admin to approve or reject a Pending listing with an optional remark. Only Approved listings shall be visible to Students.	High

PIP-F-006	The system shall allow a Recruiter to edit or close only the listings created by that Recruiter. A closed listing shall not accept new applications.	Medium
PIP-F-007	The system shall allow a Student to search Approved listings by keyword and filter by skill, location, mode and stipend range. Results shall be paginated with 10 listings per page.	High

5.3 Applications

Description and Priority: Students apply and track applications; Recruiters review and update status; notifications are sent. Priority: High.

Stimulus/Response Sequences:

- Student clicks Apply -> system validates deadline and duplicates, stores application.
- Recruiter updates status -> system saves, notifies the Student.

Functional Requirements:

ID	Requirement	Priority
PIP-F-008	The system shall allow a Student to apply to an Approved listing before its deadline using the resume in the student profile. The system shall reject duplicate applications to the same listing and applications after the deadline.	High
PIP-F-009	The system shall allow a Student to view all own applications with the current status: Submitted, Under Review, Shortlisted, Rejected or Selected.	High
PIP-F-010	The system shall allow a Recruiter to view applicants (profile and resume) of own listings and to update the application status.	High
PIP-F-011	The system shall notify a Student through an in-app notification and an email within 1 minute of any change in the status of the application.	Medium

5.4 Aptitude Tests

Description and Priority: Timed tests from a managed question bank with automatic grading and review. Priority: High.

Stimulus/Response Sequences:

- Admin adds questions/tests -> stored in question bank.
- Student starts test -> timer starts, answers autosaved.
- Student submits or time expires -> system grades and displays results.

Functional Requirements:

ID	Requirement	Priority
PIP-F-012	The system shall allow an Admin to create aptitude tests and manage a question bank. Each question shall have a category (quantitative, logical, verbal or technical), a difficulty level, options and one correct answer.	High
PIP-F-013	The system shall allow a Student to start a timed aptitude test. The system shall auto-submit the attempt when the configured duration expires.	High
PIP-F-014	The system shall auto-grade a submitted attempt and display the total score and the category-wise score to the Student immediately after submission.	High
PIP-F-015	The system shall allow a Student to review past attempts, including the correct answers, only after the attempt has been submitted. Answers shall not be sent to the client during an active attempt.	Medium

PIP-F-016	The system shall save the selected answers of an active attempt at regular intervals so that the Student can resume the attempt after a connection loss within the remaining test time.	Medium
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5.5 Progress Tracking and Analytics

Description and Priority: Dashboards and recommendations for Students and aggregate analytics for Admin. Priority: High.

Stimulus/Response Sequences:

- Student opens dashboard -> system fetches statistics from the analytics service.
- Admin opens analytics/exports report -> system displays or generates CSV.

Functional Requirements:

ID	Requirement	Priority
PIP-F-017	The system shall display a Student dashboard showing applications by status, the trend of test scores across attempts, and category-wise performance.	High
PIP-F-018	The system shall recommend practice categories to a Student based on the lowest category-wise scores of the Student, computed by the analytics service.	Medium
PIP-F-019	The system shall display aggregate analytics to an Admin: applications per listing, average test scores per category, and the number of Selected students.	Medium
PIP-F-020	The system shall allow an Admin to export registration and application reports in CSV format.	Low

5.6 Administration and Audit

Description and Priority: User management and audit logging. Priority: Medium.

Stimulus/Response Sequences:

- Admin deactivates user -> user cannot log in; audit entry written.

Functional Requirements:

ID	Requirement	Priority
PIP-F-021	The system shall allow an Admin to deactivate and reactivate user accounts. A deactivated user shall not be able to log in.	Medium
PIP-F-022	The system shall record an audit log entry (actor, action, target, timestamp) for listing approvals, application status changes and account deactivations.	Medium

6. Other Nonfunctional Requirements

6.1 Performance Requirements

ID	Requirement
PIP-NF-001	At least 95% of API requests (excluding analytics and report export) shall complete within 2 seconds with 500 concurrent users.
PIP-NF-002	A submitted test attempt shall be graded and the result returned within 3 seconds for a test of up to 50 questions.

6.2 Safety Requirements

The Portal does not control physical equipment. Safety concerns are limited to data loss and unfair test outcomes: (a) the system shall autosave test answers so a failure does not silently lose a Student's attempt (PIP-F-016); (b) data shall be backed up daily (PIP-NF-008).

6.3 Security Requirements

Security Objectives

ID	Objective	Description
SO-1	Confidentiality	Protect personal data, credentials and resumes of users from unauthorised disclosure.
SO-2	Integrity	Ensure that listings, application statuses and test scores can be changed only by authorised roles through valid operations.
SO-3	Availability	Keep the portal usable under abusive traffic such as brute-force login attempts and request flooding.
SO-4	Accountability	Ensure that security-relevant actions can be traced to an authenticated user.

Security Requirements

ID	Objective	Requirement
PIP-S-001	SO-1	All client-server and service-to-service communication shall use TLS 1.2 or higher.
PIP-S-002	SO-1	Passwords shall be stored only as bcrypt hashes with cost factor 10 or higher; plain-text passwords shall never be stored or logged.
PIP-S-003	SO-1, SO-2	The system shall enforce role-based access control (Student, Recruiter, Admin) on every API endpoint. A request outside the role's permission shall return HTTP 403.
PIP-S-004	SO-2	All user input shall be validated on the server and sanitised to prevent NoSQL injection and cross-site scripting. Invalid input shall return HTTP 400.
PIP-S-005	SO-3	The system shall limit requests to 100 per minute per IP address and shall return HTTP 429 when exceeded.
PIP-S-006	SO-1	A resume shall be accessible only to its owner, Admins, and Recruiters of the listing to which the Student applied.
PIP-S-007	SO-4	The system shall write security events (login success/failure, logout, role-violation attempts) to an append-only audit log without storing passwords or tokens.

6.4 Software Quality Attributes

ID	Attribute	Requirement
PIP-NF-003	Availability	The portal shall be available at least 99.5% of the time in a calendar month, excluding planned maintenance announced 24 hours in advance.
PIP-NF-004	Usability	A first-time Student shall be able to register, complete the profile and submit a first application within 10 minutes without training. The UI shall be responsive from 360 px screen width.
PIP-NF-005	Compatibility	The web application shall work on the latest two stable versions of Chrome, Firefox and Edge.
PIP-NF-006	Scalability	The system shall support 5,000 registered users and 2,000 test attempts per day, and the application tier shall be horizontally scalable.

PIP-NF-007	Maintainability	Backend modules shall have at least 70% unit-test line coverage and shall pass the ESLint (JavaScript) and flake8 (Python) checks without errors.
PIP-NF-008	Reliability	The database shall be backed up daily and restorable with a data loss of at most 24 hours.

Relative preference: security and correctness of test grading take priority over visual polish; ease of use is preferred over ease of learning.

6.5 Business Rules

- BR-1: Only Admins can approve or reject listings.
- BR-2: Only the Recruiter who created a listing can edit, close or review its applicants.
- BR-3: A Student can have only one application per listing.
- BR-4: A test attempt is final once submitted; it cannot be retaken within the same attempt.
- BR-5: Only Approved, open listings before the deadline accept applications.

Domain Requirements

- Application statuses follow the order Submitted -> Under Review -> Shortlisted/Rejected -> Selected.
- Aptitude categories are Quantitative, Logical, Verbal and Technical.
- Personal data handling shall follow the principles of the Digital Personal Data Protection Act, 2023 (consent, purpose limitation, minimal retention).

7. Other Requirements

- Database: MongoDB collections with indexes on listing search fields and (studentId, listingId) uniqueness.
- Legal: users shall accept terms and privacy policy at registration.
- Internationalisation: English only in version 1.0.
- Reuse: validation and auth middleware shall be shared across API modules.

Appendix A: Glossary

Term	Meaning
Portal	Online Internship and Placement Preparation Portal
MERN	MongoDB, Express.js, React, Node.js
JWT	JSON Web Token used for session tokens
RBAC	Role-Based Access Control
SRS / SAD / STP	Software Requirements Specification / Software Architecture and Design / Software Test Plan
TLS	Transport Layer Security
CSV	Comma-Separated Values
REST	Representational State Transfer
Attempt	One sitting of a Student on an aptitude test
Listing	An internship posting by a Recruiter

Appendix B: Field Layouts

Registration fields (User)

Field	Length	Data Type	Description	Mandatory
Name	60	String	Full name of the user	Y
Email	100	String	Unique login identifier	Y
Password	8-64	String	Stored as bcrypt hash	Y
Role	10	Enum	Student / Recruiter	Y
Phone	15	Numeric	Contact number	N

Listing fields

Field	Length	Data Type	Description	Mandatory
Title	100	String	Internship title	Y
Company	80	String	Company name	Y
Description	2000	String	Role details	Y
Skills	-	String list	Required skills	Y
Stipend	8	Numeric	Monthly stipend	N
Mode	10	Enum	Remote / Onsite / Hybrid	Y
Deadline	10	Date	Last date to apply	Y
Status	10	Enum	Pending / Approved / Rejected / Closed	Y

Report requirements

Registration Report	Application Report
User ID	Application ID
Name	Student Name
Email	Listing Title
Role	Company
Registration Date	Applied Date
Account Status	Status

Appendix C: Requirement Traceability Matrix

Architecture and design references point to the SAD sections; test case IDs point to the Software Test Plan.

SI	Req. ID	Brief Description	Architecture Ref.	Design Ref.	Code File Ref.	Test Case ID	System Test
1	PIP-F-001	The system shall allow a visitor to register as a Student or Recruiter by providing name, ...	Auth & User Service (SAD 3.4)	SAD 4.3 Auth API	server/modules/auth, client/pages/auth	TC-AUTH-01, TC-AUTH-02	System test
2	PIP-F-002	The system shall authenticate users using email and password and issue a session token val...	Auth & User Service (SAD 3.4)	SAD 4.3 Auth API	server/modules/auth, client/pages/auth	TC-AUTH-03	System test
3	PIP-F-003	The system shall allow a Student to create and edit a profile containing education, skills...	Auth & User Service (SAD 3.4)	SAD 4.3 Auth API	server/modules/auth, client/pages/auth	Unit / integration tests	System test
4	PIP-F-004	The system shall allow a Recruiter to create an internship listing with title, company, de...	Internship Service (SAD 3.4)	SAD 4.3 Internship API	server/modules/internships	TC-INT-01	System test

5	PIP-F-005	The system shall allow an Admin to approve or reject a Pending listing with an optional re...	Internship Service (SAD 3.4)	SAD 4.3 Internship API	server/module s/internships	TC-INT-01	System test
6	PIP-F-006	The system shall allow a Recruiter to edit or close only the listings created by that Recr...	Internship Service (SAD 3.4)	SAD 4.3 Internship API	server/module s/internships	Unit / integration tests	System test
7	PIP-F-007	The system shall allow a Student to search Approved listings by keyword and filter by skil...	Internship Service (SAD 3.4)	SAD 4.3 Internship API	server/module s/internships	TC-INT-02	System test
8	PIP-F-008	The system shall allow a Student to apply to an Approved listing before its deadline using...	Application Service, Notification Service (SAD 3.4)	SAD 4.2 Seq. 1; 4.3 Application API	server/module s/applications	TC-APP-01, TC-APP-02	System test
9	PIP-F-009	The system shall allow a Student to view all own applications with the current status: Sub...	Application Service, Notification Service (SAD 3.4)	SAD 4.2 Seq. 1; 4.3 Application API	server/module s/applications	TC-APP-01	System test
10	PIP-F-010	The system shall allow a Recruiter to view applicants (profile and resume) of own listings...	Application Service, Notification Service (SAD 3.4)	SAD 4.2 Seq. 1; 4.3 Application API	server/module s/applications	TC-APP-03	System test
11	PIP-F-011	The system shall notify a Student through an in-app notification and an email within 1 min...	Application Service, Notification Service (SAD 3.4)	SAD 4.2 Seq. 1; 4.3 Application API	server/module s/applications	TC-APP-03	System test
12	PIP-F-012	The system shall allow an Admin to create aptitude tests and manage a question bank	Aptitude Test Service (SAD 3.4)	SAD 4.2 Seq. 2; 4.3 Test API	server/module s/tests	Unit / integration tests	System test
13	PIP-F-013	The system shall allow a Student to start a timed aptitude test	Aptitude Test Service (SAD 3.4)	SAD 4.2 Seq. 2; 4.3 Test API	server/module s/tests	TC-TST-01	System test
14	PIP-F-014	The system shall auto-grade a submitted attempt and display the total score and the catego...	Aptitude Test Service (SAD 3.4)	SAD 4.2 Seq. 2; 4.3 Test API	server/module s/tests	TC-TST-02	System test
15	PIP-F-015	The system shall allow a Student to review past attempts, including the correct answers, o...	Aptitude Test Service (SAD 3.4)	SAD 4.2 Seq. 2; 4.3 Test API	server/module s/tests	TC-TST-02	System test
16	PIP-F-016	The system shall save the selected answers of an active attempt at regular intervals so th...	Aptitude Test Service (SAD 3.4)	SAD 4.2 Seq. 2; 4.3 Test API	server/module s/tests	TC-TST-01	System test
17	PIP-F-017	The system shall display a Student dashboard showing applications by status, the trend of ...	Analytics Service (SAD 3.4)	SAD 4.3 Analytics API	analytics/app, client/pages/dashboard	TC-ANL-01	System test
18	PIP-F-018	The system shall recommend practice categories to a Student based on the lowest category-w...	Analytics Service (SAD 3.4)	SAD 4.3 Analytics API	analytics/app, client/pages/dashboard	TC-ANL-01	System test
19	PIP-F-019	The system shall display aggregate analytics to an Admin: applications per listing, averag...	Analytics Service (SAD 3.4)	SAD 4.3 Analytics API	analytics/app, client/pages/dashboard	Unit / integration tests	System test
20	PIP-F-020	The system shall allow an Admin to export registration and application reports in CSV form...	Analytics Service (SAD 3.4)	SAD 4.3 Analytics API	analytics/app, client/pages/dashboard	Unit / integration tests	System test
21	PIP-F-021	The system shall allow an Admin to deactivate and reactivate user accounts	Admin & Report Service (SAD 3.4)	SAD 4.4 Logging	server/module s/admin	Unit / integration tests	System test
22	PIP-F-022	The system shall record an audit log entry (actor, action, target, timestamp) for listing ...	Admin & Report Service (SAD 3.4)	SAD 4.4 Logging	server/module s/admin	Unit / integration tests	System test
23	PIP-NF-001	At least 95% of API requests (excluding analytics and report export) shall complete within...	Cross-cutting (SAD 3.1, 3.6)	-	All modules	TC-PERF-01	System test

24	PIP-NF-002	A submitted test attempt shall be graded and the result returned within 3 seconds for a te...	Cross-cutting (SAD 3.6)	-	All modules	TC-PERF-02	System test
25	PIP-NF-003	The portal shall be available at least 99.5% of the time in a calendar month, excluding pl...	Cross-cutting (SAD 3.7)	-	All modules	Monitoring and review (STP 5)	System test
26	PIP-NF-004	A first-time Student shall be able to register, complete the profile and submit a first ap...	Cross-cutting (SAD 4.5)	-	All modules	TC-USA-01	System test
27	PIP-NF-005	The web application shall work on the latest two stable versions of Chrome, Firefox and Ed...	Cross-cutting (SAD 4.5)	-	All modules	TC-USA-01	System test
28	PIP-NF-006	The system shall support 5,000 registered users and 2,000 test attempts per day, and the a...	Cross-cutting (SAD 3.5)	-	All modules	Monitoring and review (STP 5)	System test
29	PIP-NF-007	Backend modules shall have at least 70% unit-test line coverage and shall pass the ESLint ...	Cross-cutting (SAD 3.5)	-	All modules	Monitoring and review (STP 5)	System test
30	PIP-NF-008	The database shall be backed up daily and restorable with a data loss of at most 24 hours....	Cross-cutting (SAD 3.7)	-	All modules	Monitoring and review (STP 5)	System test
31	PIP-S-001	All client-server and service-to-service communication shall use TLS 1.2 or higher.	Security Architecture (SAD 3.9)	SAD 4.4	middleware/*	TC-SEC-03	System test
32	PIP-S-002	Passwords shall be stored only as bcrypt hashes with cost factor 10 or higher; plain-text ...	Security Architecture (SAD 3.9)	SAD 4.4	middleware/*	TC-SEC-03	System test
33	PIP-S-003	The system shall enforce role-based access control (Student, Recruiter, Admin) on every AP...	Security Architecture (SAD 3.9)	SAD 4.4	middleware/*	TC-SEC-01	System test
34	PIP-S-004	All user input shall be validated on the server and sanitised to prevent NoSQL injection a...	Security Architecture (SAD 3.9)	SAD 4.4	middleware/*	TC-SEC-02	System test
35	PIP-S-005	The system shall limit requests to 100 per minute per IP address and shall return HTTP 429...	Security Architecture (SAD 3.9)	SAD 4.4	middleware/*	TC-SEC-03	System test
36	PIP-S-006	A resume shall be accessible only to its owner, Admins, and Recruiters of the listing to w...	Security Architecture (SAD 3.9)	SAD 4.4	middleware/*	TC-SEC-01	System test
37	PIP-S-007	The system shall write security events (login success/failure, lockout, role-violation att...	Security Architecture (SAD 3.9)	SAD 4.4	middleware/*	TC-SEC-03	System test